Integer Multiplication as Convolution

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Abstract

Finite sequences of nonnegative integers can be packed into the digits, or fixed-width digit blocks, of ordinary integers. If the packing base is large enough to prevent carries, multiplying the packed integers computes the exact linear convolution of the original sequences. This elementary construction is a decimal instance of *Kronecker substitution*, a standard method for reducing polynomial multiplication to integer multiplication.

1 The basic idea

Let

$$a = (a_0, a_1, \ldots, a_m), \qquad b = (b_0, b_1, \ldots, b_n)$$

be finite sequences of nonnegative integers. Their linear convolution is the sequence $c = a * b$ defined by

$$c_k = \sum_{\substack{i+j=k \\ 0\le i\le m,\ 0\le j\le n}} a_i b_j, \qquad 0 \le k \le m+n.$$

Choose an integer base $B \ge 2$ and form the packed integers

$$A = \sum_{i=0}^{m} a_i B^i, \qquad D = \sum_{j=0}^{n} b_j B^j.$$

Multiplication and collection of equal powers of B give

$$\begin{aligned} AD &= \left(\sum_{i=0}^{m} a_i B^i\right)\left(\sum_{j=0}^{n} b_j B^j\right) \\ &= \sum_{k=0}^{m+n} \left(\sum_{i+j=k} a_i b_j\right) B^k \\ &= \sum_{k=0}^{m+n} c_k B^k. \end{aligned}$$

Consequently, if every $c_k < B$, the base-B digits of AD, read from least significant to most significant, are exactly the convolution coefficients $c_0, c_1, \ldots, c_{m+n}$. There are no carries from one coefficient block to the next.

Equivalently, associate the polynomials

$$p(x) = \sum_{i=0}^{m} a_i x^i, \qquad q(x) = \sum_{j=0}^{n} b_j x^j.$$

The coefficient sequence of $p(x)q(x)$ is $a * b$, while the packed integers are simply $p(B)$ and $q(B)$. The identity

$$p(B)q(B) = (pq)(B)$$

is therefore the entire mechanism.

2 A decimal example

For

$$a = b = (1, 1, 1),$$

we have

$$p(x) = q(x) = 1 + x + x^2.$$

Packing in base 10 gives $p(10) = q(10) = 111$, and hence

$$111^2 = 12321.$$

Reading the decimal digits from either direction in this symmetric example gives

$$(1, 1, 1) * (1, 1, 1) = (1, 2, 3, 2, 1).$$

More generally, the convolution of two all-ones sequences of lengths r and s rises linearly to $\min(r, s)$, may have a plateau, and then falls linearly. Base 10 displays those coefficients directly only while all of them are at most 9.

The order convention deserves mention. In the definition above, a_0 is the least significant base-B digit. Thus the written numeral normally displays the sequence in reverse order. The example 111 hides this point because it is palindromic.

3 Carries and the choice of base

The construction is exact only when the base is larger than every possible convolution coefficient. Suppose

$$0 \le a_i \le M, \qquad 0 \le b_j \le N.$$

Each c_k is a sum of at most $\min(m + 1, n + 1)$ products, and therefore

$$c_k \le \min(m + 1, n + 1)MN.$$

It is sufficient to choose

$$B > \min(m + 1, n + 1)MN.$$

This bound is sometimes conservative, but it is immediate and convenient.

For example, consider

$$a = (1, 2), \qquad b = (3, 4).$$

Their convolution is

$$a * b = (3, 10, 8).$$

Base 10 is too small: the encodings 21 and 43 give

$$21 \cdot 43 = 903,$$

and the carry from the middle coefficient 10 obscures the desired answer. This does not mean that multiplication has ceased to involve convolution; rather, ordinary digit carrying has normalized the raw convolution coefficients into legal base-10 digits.

4 Base 100 and fixed-width decimal blocks

Using base 100 means treating each consecutive pair of decimal digits as one coefficient block. For the preceding example,

$$A = 1 + 2(100) = 201, \qquad D = 3 + 4(100) = 403.$$

Then

$$201 \cdot 403 = 81003.$$

After padding the leading block to two places, the product is

$$08 \mid 10 \mid 03.$$

Reading the blocks from least significant to most significant gives

$$(03, 10, 08) = (3, 10, 8),$$

the exact convolution.

More generally, base 10^d stores one coefficient in each block of d decimal places. One chooses d so that

$$10^d > \max_k c_k$$

or uses an a priori bound such as the one above. For the self-convolution of an all-ones sequence of length r, the largest coefficient is r. Therefore base 10 works for $r \leq 9$, base 100 works for $r \leq 99$, and base 10^d works for $r < 10^d$.

Thus base 100 does allow larger entries and longer sequences than base 10, but the decisive quantity is the size of the *output* coefficients, not merely the sizes of the input entries. In norm notation, the elementary estimate

$$\|a * b\|_\infty \leq \min(m + 1, n + 1)\|a\|_\infty \|b\|_\infty$$

provides a safe packing base.

5 Relation to standard terminology

This method is usually called *Kronecker substitution*: a polynomial with integer coefficients is evaluated at a sufficiently large integer base, polynomial multiplication is replaced by one integer multiplication, and the product coefficients are recovered from the base-B representation. The reverse unpacking operation is also called *Kronecker segmentation*. In implementations, the same idea is often described as *coefficient packing*, generally using bases that are powers of 2 rather than powers of 10.

The construction also explains a familiar fact about long multiplication. Before carries are performed, multiplying two digit strings produces the discrete convolution of their digit sequences. Carrying converts those raw coefficients into the standard positional representation. Conversely, if the base is deliberately made larger than every raw coefficient, no carrying is necessary and the convolution remains visible in the product.

This should be distinguished from multiplication that is intrinsically *carryless*, where coefficients may instead be reduced in a specified coefficient ring. The present construction performs ordinary integer convolution over the nonnegative integers and prevents carries by allocating enough space to each coefficient.

Signed coefficients can also be handled, but their encoding and decoding require additional conventions, such as balanced digits, offsets, or separate positive and negative parts. The nonnegative case is the cleanest pedagogical form.

6 Historical and pedagogical context

The relationship among digit multiplication, polynomial multiplication, and convolution is standard. Smith's text on the mathematics of the discrete Fourier transform explains directly that multiplication convolves the digit sequences, with carrying as the extra step in ordinary arithmetic [**?**]. Sutherland's lecture notes present the converse reduction under the heading "Kronecker substitution": polynomial coefficients are packed into separated binary blocks, the resulting integers are multiplied, and the coefficients are recovered from the product [**?**].

A systematic computer-algebra treatment appears in von zur Gathen and Gerhard's *Modern Computer Algebra* [**?**]. Kronecker substitution is important computationally because it permits polynomial arithmetic to use highly optimized large-integer multiplication. Modern research continues to use and refine this reduction; for example, Harvey, van der Hoeven, and Lecerf use Kronecker substitution and segmentation within fast integer-multiplication algorithms [**?**], while Harvey studies a multipoint variant for dense polynomial multiplication [**?**].

The particular display

$$111^2 = 12321$$

is often presented as a recreational number pattern. Interpreting it as the self-convolution of the finite sequence ((1,1,1)) supplies the structural explanation: it is the coefficient sequence of $(1 + x + x^2)^2$, evaluated at $x = 10$. The same explanation immediately shows both why the pattern works and why it eventually encounters carries.

References

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